Economic Applications of Game Theory

Personal Notes

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1 Introduction

Game theory is a misnomer for *multiperson decision theory*: it develops tools, methods, and language for analyzing the choices of agents whose payoffs depend on each other's actions. A player's preference over his own actions depends on his beliefs about what the others do; what the others do depends on their beliefs about what each player does; and so on, ad infinitum. This chapter illustrates the main reasoning moves of the course on a handful of small games. The names will return in later chapters with full definitions; here the point is to see them at work.

1.1 A Game Solved by Iterated Reasoning

Throughout, payoffs are written as ordered pairs (u_1, u_2): the first entry is Player 1's payoff, the second is Player 2's. Player 1 chooses the row, Player 2 chooses the column, and they choose simultaneously. Consider the following game.

Player 1 / Player 2	L	m	R
T	$(1,1)$	$(0,2)$	$(2,1)$
M	$(2,2)$	$(1,1)$	$(0,0)$
B	$(1,0)$	$(0,0)$	$(-1,1)$

Assume each player knows the strategies and payoffs, each knows that the other knows, and so on without limit — call this *common knowledge* of the game.

Player 1 compares M and B. For every column, M pays more: $2 > 1$ under L, $1 > 0$ under m, and $0 > -1$ under R. So B is worse than M no matter what Player 2 does, and Player 1 should not play B. None of L, m, R is initially unambiguously best for Player 2: R beats m if and only if Player 1 plays B, and otherwise m beats R. But Player 2 can rule out B. With B gone, m beats R everywhere, so R is ruled out. Player 1 then knows Player 2 rules out R, and in the remaining 2×2 game M beats T. Player 1 plays M, and Player 2, expecting M, plays L. The predicted outcome is (M, L) with payoff $(2, 2)$.

Note: *Iterated elimination repeatedly removes a strategy that is strictly worse than another strategy against every remaining action of the opponent. In this game the elimination order is B, then R, then T, leaving (M, L).*

Exercise. The reasoning above assumes a great deal about what each player assumes about the other's reasoning. Identify those assumptions. How would the analysis change if players act rationally but each assumes the other plays a uniformly random strategy?

1.2 When Iterated Reasoning Gives No Answer: Coordination

The previous game admits a sharp prediction. Most games do not. Suppose you want to meet a friend in one of two equally good places, and you cannot communicate. This is the *pure coordination game*.

Player 1 / Player 2	Left	Right
Top	$(1,1)$	$(0,0)$
Bottom	$(0,0)$	$(1,1)$

Player 1 prefers Top if he expects Left and Bottom if he expects Right; symmetrically for Player 2. Iterated elimination removes nothing. Asking "what does the other do?" recurses without progress.

One way out is to ask for *stable* profiles — profiles such that no player wants to deviate given the others' play.

Definition 1.1 (Nash equilibrium)**.** A strategy profile $(s_1^*, s_2^*, \ldots)$ is a Nash equilibrium if, for each player i, s_i^* maximizes player i's payoff given that the other players play s_{-i}^*. Equivalently, no player has a profitable unilateral deviation.

In the pure coordination game both (Top, Left) and (Bottom, Right) are Nash equilibria. The two unmatched profiles are not: in each case, both players want to switch.

1.2.1 Battle of the Sexes

Usually preferences over outcomes disagree as well, and coordination must happen against a conflict. The classical *Battle of the Sexes* mixes the two.

Player 1 / Player 2	Left	Right
Top	$(2,1)$	$(0,0)$
Bottom	$(0,0)$	$(1,2)$

Both players want to coordinate, but Player 1 prefers (Top, Left) while Player 2 prefers (Bottom, Right). The Nash equilibria are again the two matched cells.

1.3 Sequential Moves and Backward Induction

So far the players have moved simultaneously. If one player observes the other's action before choosing, the structure of the game — and its predictions — can change. Imagine the Battle of the

Sexes played sequentially: Player 1 commits first, Player 2 sees the move, then Player 2 chooses. In the tree below the strategies are written B and O ("Bach" and "Opera"); B corresponds to Top/Left in the simultaneous matrix, and O to Bottom/Right.

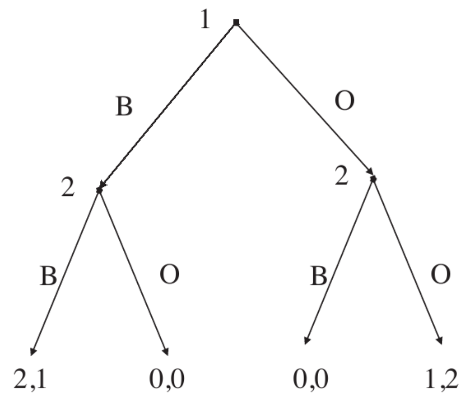


Figure 1.1: Battle of Sexes with sequential moves

Solve from the leaves up. If Player 1 played B, Player 2 picks B for a payoff of 1 instead of 0. If Player 1 played O, Player 2 picks O for 2 instead of 0. Anticipating this, Player 1 plays B — payoff 2 rather than 1 — and the outcome is (B, B) with payoffs $(2, 1)$. This solution method is called *backward induction*: work from the terminal nodes upward, replacing each subtree by its predicted payoff.

In the simultaneous Battle of the Sexes there were two equilibria, giving Player 2 either 1 or 2. In the sequential version, Player 2 gets 1 for sure. Giving Player 2 better information about Player 1's move *hurt* Player 2. When it is common knowledge that a player has some piece of information, the player may prefer not to have it. The general phenomenon — information can hurt — recurs throughout the course.

Exercise. Suppose Player 2 sees Player 1's move only with probability $\pi < 1$, independent of what Player 1 chose. What is a reasonable equilibrium? By the end of the course, the model will let us formalize and compute it.

A second reading of the sequential game is that Player 1 can *commit* in a way Player 2 cannot. Commitment is power. Forcing yourself to act first, while the other side merely reacts, can be worth more than retaining flexibility.

Exercise. Extend the sequential game so that, after Player 2 chooses, Player 1 gets a chance to switch his action; then the game ends. What is the reasonable outcome? What if switching costs Player 1 c utiles?

1.4 Forward Induction

Suppose that before playing the Battle of the Sexes, Player 1 may take an outside option that pays $(3/2, 3/2)$ to both players. If he plays, Player 2 sees that he chose to play and they enter the 2×2 Battle of the Sexes.

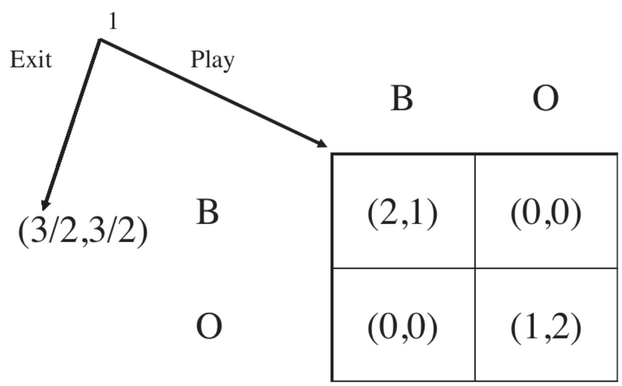


Figure 1.2: Battle of Sexes with exit option

Two stable patterns of play exist. In the first, Player 1 exits because he expects Player 2 to insist on (O, O) inside the subgame, leaving him with only 1, so 3/2 is better. In the second, Player 1 enters and they coordinate on (B, B), giving him 2.

The first outcome is questionable. When Player 2 is asked to move inside the subgame, she *knows* Player 1 gave up the certain 3/2. She must conclude he is not aiming at the (O, O) cell, which pays him only 1. So he must be aiming at (B, B), and she should play B. Anticipating this, Player 1 enters and they coordinate on (B, B). This style of reasoning is called *forward induction*: a deviation off the supposed equilibrium path is read as a signal about how the deviator intends to play later.

1.5 Three Games That Recur

1.5.1 Prisoners' Dilemma

Player 1 / Player 2	Cooperate	Defect
Cooperate	$(5, 5)$	$(0, 6)$
Defect	$(6, 0)$	$(1, 1)$

Two suspects, two rooms, no firm evidence. Each can keep quiet (cooperate) or confess (defect). For each player, Defect strictly beats Cooperate against either action of the opponent, so both defect and the outcome is $(1, 1)$ — even though both prefer $(5, 5)$. Individually rational play makes both worse off.

1.5.2 Hawk–Dove

Player 1 / Player 2	Hawk	Dove
Hawk	$\left(\frac{V-C}{2}, \frac{V-C}{2}\right)$	$(V, 0)$
Dove	$(0, V)$	$\left(\frac{V}{2}, \frac{V}{2}\right)$

Two animals contest a resource of value V. *Hawk* fights; *Dove* yields. Two hawks meeting share a fight of cost C, so each gets $(V-C)/2$. A hawk meeting a dove takes the whole resource; two doves split it.

If $V > C$, fighting is cheap, Hawk strictly dominates Dove, and the game is essentially a Prisoners' Dilemma. If $V < C$, fighting is costly, and each player wants Hawk against a Dove and Dove against a Hawk; the game becomes a coordination problem with conflict, structurally a version of Chicken and the prototype for *wars of attrition*.

1.5.3 Investment Game

Player 1 / Player 2	Invest	Do not invest
Invest	(θ, θ)	$(\theta - c, 0)$
Do not invest	$(0, \theta - c)$	$(0, 0)$

Two parties decide simultaneously whether to invest at cost c. Investment pays θ when matched and $\theta - c$ when not. Think of a worker considering education and a firm considering a technology that requires educated workers. The reasonable outcomes depend on the relative sizes of θ and c — and on what each side knows about the other's parameters. Information about the opponent's payoff matters as much as the payoff itself, a theme the second half of the course makes precise.

Source: MIT 14.12 lecture notes by Muhamet Yildiz.

2 Decision Theory

The first lecture treated payoffs as given. This chapter explains what those payoffs mean. A game-theoretic payoff is not merely a ranking of final outcomes; it is a compact representation of how a player compares uncertain consequences induced by beliefs about the other players' actions.

2.1 Choice Under Certainty

Start with a set X of mutually exclusive, exhaustive alternatives. A decision maker's weak preference relation $\geq$ compares alternatives: $x \geq y$ means that x is at least as good as y.

Definition 2.1 (Preference relation)**.** A relation $\succeq$ on X is a preference relation if it is complete and transitive. Completeness says that for every $x, y \in X$, either $x \succeq y$ or $y \succeq x$. Transitivity says that $x \succeq y$ and $y \succeq z$ imply $x \succeq z$.

Definition: For alternatives $x, y \in X$, strict preference and indifference are derived from weak preference by

$$x \succ y \iff [x \succeq y \text{ and not } y \succeq x], \qquad x \sim y \iff [x \succeq y \text{ and } y \succeq x].$$

A utility function assigns numbers to alternatives so that better alternatives receive larger numbers.

Theorem 2.2 (Utility representation under certainty)**.** *If X is finite, a relation on X can be represented by a utility function $u : X \to \mathbb{R}$,*

$$x \succeq y \iff u(x) \geq u(y),$$

if and only if the relation is complete and transitive. If u represents $\succeq$ and $f : \mathbb{R} \to \mathbb{R}$ is strictly increasing, then $f \circ u$ represents the same preferences.

The last sentence is the key. Utility under certainty is *ordinal*: only the order matters. The numerical gaps between utility values carry no independent meaning.

2.2 Decision Making Under Uncertainty

Let Z be a finite set of prizes. A *lottery* is a probability distribution $p : Z \to [0, 1]$ with $\sum_{z \in Z} p(z) = 1$. For example, the lottery below gives 10 with probability 1/2 and 0 with probability 1/2.

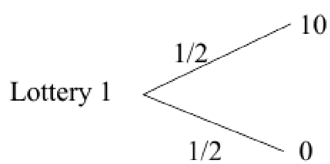


Figure 2.1:

In games, players usually do not know what the others will do. If a player has beliefs about the others' actions, each of his own actions induces a lottery over final outcomes. This is why game theory needs a theory of preference over lotteries, not just over sure prizes.

Definition 2.3 (von Neumann–Morgenstern utility)**.** A preference relation $\succeq$ over lotteries P is represented by a VNM utility function $u : Z \to \mathbb{R}$ if, for every $p, q \in P$,

$$p \succeq q \iff U(p) \equiv \sum_{z \in Z} u(z)p(z) \geq \sum_{z \in Z} u(z)q(z) \equiv U(q).$$

This definition has two layers. First, U is an ordinary utility representation of preferences over lotteries. Second, it has the special expected-utility form: the utility of a lottery is the probability-weighted average of utility over prizes. For the lottery above, $U = \frac{1}{2}u(10) + \frac{1}{2}u(0)$.

2.2.1 The Three Axioms

The VNM representation is not automatic. It follows from three assumptions about preferences over lotteries.

Assumption 2.4 (Completeness and transitivity)**.** Preferences over lotteries are complete and transitive.

Assumption 2.5 (Independence)**.** For any lotteries p, q, r and any $\alpha \in (0, 1]$,

$$\alpha p + (1 - \alpha)r \succ \alpha q + (1 - \alpha)r \iff p \succ q.$$

Independence says that mixing both options with the same background lottery should not reverse the comparison. One interpretation is dynamic consistency: if the decision maker prefers p to q, he should still prefer "first toss a coin, then get p on heads and r on tails" to the corresponding compound lottery with q replacing p.

Assumption 2.6 (Continuity)**.** If $p \succ q$, then for any third lottery r there are $\alpha, \beta \in (0, 1)$ such that

$$\alpha p + (1 - \alpha)r \succ q \quad \text{and} \quad p \succ \beta q + (1 - \beta)r.$$

Continuity rules out infinitely good or infinitely bad prizes. It is mostly technical, but it makes the utility representation finite and well behaved.

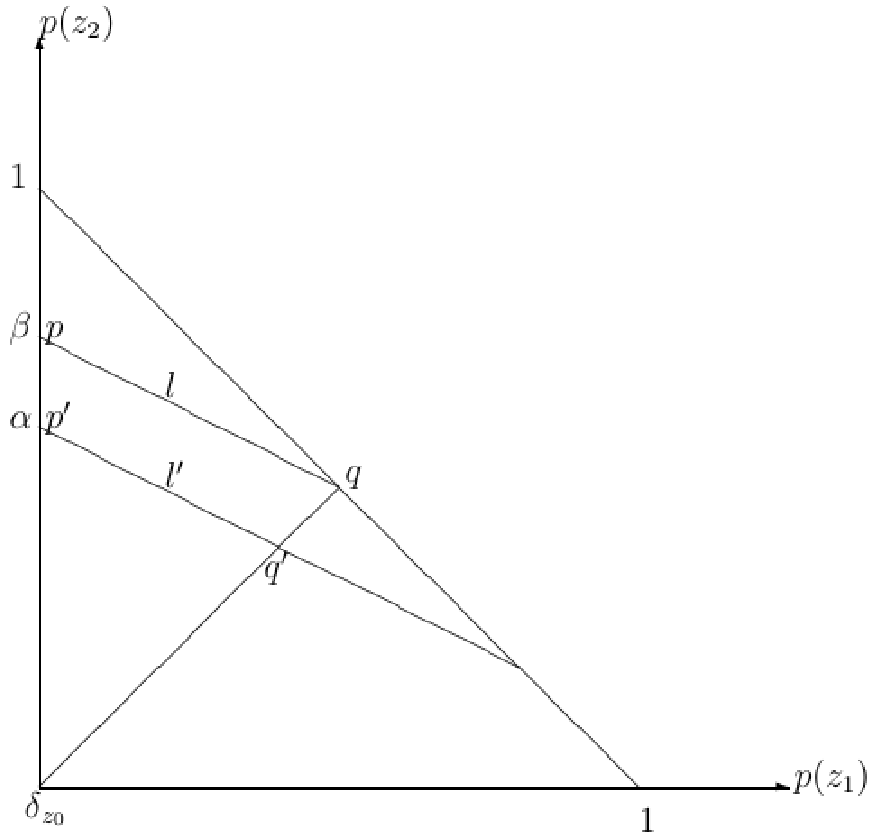
Independence has a geometric implication. If $p \sim q$, then for any r and $\alpha \in [0, 1]$,

$$\alpha p + (1 - \alpha)r \sim \alpha q + (1 - \alpha)r.$$

With three prizes, lotteries can be drawn as points in a triangle. The implication above makes indifference curves straight, and repeating the argument from different base lotteries makes them parallel.

Theorem 2.7 (VNM representation)**.** *A relation $\succeq$ on the set of lotteries P has a VNM representation if and only if it satisfies completeness and transitivity, independence, and continuity. Moreover, two VNM utility functions u and $\widetilde{u}$ represent the same preferences if and only if $\widetilde{u} = \alpha u + \beta$ for some $\alpha > 0$ and $\beta \in \mathbb{R}$.*

VNM utility is therefore *cardinal* in a limited sense. Positive affine transformations preserve preferences over lotteries; arbitrary increasing transformations need not. This is the difference between saying "only the order of sure prizes matters" and saying "utility gaps matter when computing expected utility."



2.3 Modeling Strategic Situations

Consider two players, Alice and Bob, with strategy sets S_A and S_B. If Alice plays s_A and Bob plays s_B, the outcome is (s_A, s_B). Alice does not know Bob's strategy when choosing, so a belief μ_A over S_B turns each Alice strategy into a lottery over outcomes.

Under the VNM assumptions, we do not need to list Alice's preference over every possible lottery separately. We can summarize her preferences by a payoff function

$$u_A : S_A \times S_B \to \mathbb{R},$$

and similarly for Bob. This is the mathematical justification for putting numbers in payoff matrices.

Example: Let $S_A = \{T, B\}$ and $S_B = \{L, R\}$. If Alice believes Bob plays L with probability p and R with probability $1 - p$, then T gives the lottery

$$(T, L) \text{ with probability } p, \qquad (T, R) \text{ with probability } 1 - p,$$

while B gives the corresponding lottery over (B, L) and (B, R). The belief p is not fixed by the model; equilibrium analysis will later determine which beliefs and actions are consistent.

Example: Suppose Alice prefers T to B exactly when $p > 1/4$, is indifferent at $p = 1/4$, and prefers B when $p < 1/4$. Suppose also that she is indifferent between (B, L) and (B, R). A VNM representation is

$$u_A(T, L) = 3, \quad u_A(T, R) = -1, \quad u_A(B, L) = 0, \quad u_A(B, R) = 0.$$

Indeed, $U_A(T) = 3p - (1 - p) = 4p - 1$, while $U_A(B) = 0$. The sign of $4p - 1$ changes exactly at $p = 1/4$.

Alice	L	R
T	3	-1
B	0	0

Alice's VNM payoff representation in the preceding example.

2.4 Attitudes Toward Risk

Now specialize prizes to money and write $u : \mathbb{R} \to \mathbb{R}$. A fair gamble has expected monetary value 0. A risk-neutral decision maker is indifferent between accepting and rejecting every fair gamble, which happens exactly when u is linear. A strictly risk-averse decision maker rejects every nontrivial fair gamble, which corresponds to strictly concave u. Risk seeking corresponds to convex u.

Definition 2.8 (Attitudes toward risk)**.** For monetary lotteries, a VNM utility function is

- *risk-neutral* if u is linear;
- *risk-averse* if u is concave;
- *risk-seeking* if u is convex.

For a concave utility function, the utility of expected wealth lies above the expected utility of risky wealth:

$$u(pW_1 + (1 - p)W_2) \geq p\, u(W_1) + (1 - p)u(W_2).$$

The vertical gap is the utility cost of risk. In money terms, it becomes the most the person would pay to replace the risky gamble with a sure amount.

2.4.1 Risk Sharing

Let $u(x) = \sqrt{x}$. A person owns an asset that pays 100 with probability 1/2 and 0 with probability 1/2. The expected utility is

$$\frac{1}{2}\sqrt{100} + \frac{1}{2}\sqrt{0} = 5.$$

If two identical people pool independent copies of this asset and each owns half the pool, each person's share pays 100 with probability 1/4, 50 with probability 1/2, and 0 with probability 1/4.

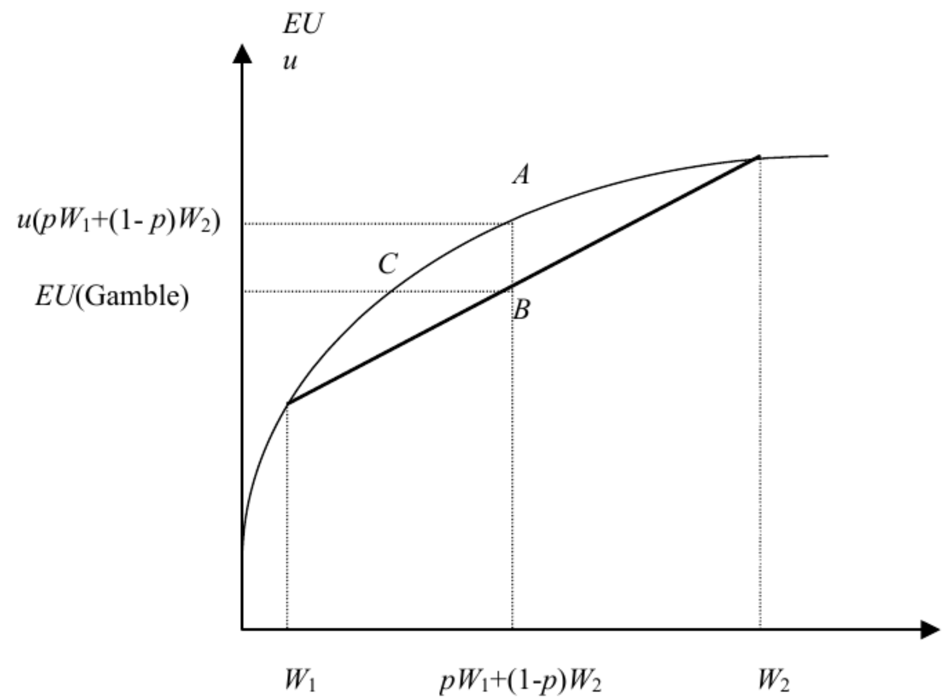


Figure 2.5:

Expected utility becomes

$$\frac{1}{4}\sqrt{100} + \frac{1}{2}\sqrt{50} + \frac{1}{4}\sqrt{0} \approx 6.0355.$$

Risk sharing raises expected utility because the pool smooths each person's payoff.

2.4.2 Insurance

Keep the same risk-averse agent and add a risk-neutral insurer. Full insurance pays 100 exactly when the asset pays 0. If the premium is P, the insured person's wealth is $100 - P$ for sure. He buys insurance when

$$\sqrt{100 - P} \geq \frac{1}{2}\sqrt{100} + \frac{1}{2}\sqrt{0} = 5,$$

so $P \leq 75$. The risk-neutral insurer receives P for sure and expects to pay 50, so it is willing to sell when $P \geq 50$. Any premium in $(50, 75)$ benefits both sides.

Note: For a high payoff H realized with probability $1/2$, the uninsured expected utility is $\sqrt{H}/2$. Full insurance is acceptable to the buyer when the premium does not exceed $3H/4$, while a risk-neutral insurer requires at least $H/2$. The interval $(H/2, 3H/4)$ therefore benefits both parties.

Exercise. Suppose two identical risk-averse agents can either buy insurance at a common premium or first form the mutual fund above. What range of premiums can make every party willing to participate?

2.5 Takeaways

Payoff numbers in a game are VNM utility numbers. They summarize how a player evaluates lotteries over outcomes, and they are unique only up to positive affine transformations. Replacing payoffs by an arbitrary increasing transformation preserves rankings of sure outcomes but can change preferences under uncertainty, and therefore can change the game.

Source: MIT 14.12 lecture notes by Muhamet Yildiz.